Automated Road Pothole Detection and Spatial Severity Auditing via Hand-Crafted Feature Fusion and Deep Representations

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Abstract*—Potholes and structural road degradation present severe vehicular hazards, escalate maintenance expenditure, and compromise civil transport safety. Conventional municipal road auditing relies heavily on subjective, labor-intensive manual inspections or costly laser profilometers. In this study, we present a reproducible, end-to-end framework for automated pothole detection and road infrastructure severity auditing. Using a real-world multi-national benchmark of 15,340 driving images spanning three continents from the Road Damage Dataset (RDD2022)—Czech Republic, India, and the United States—we design a 10,217-dimensional hand-crafted descriptor fusing HSV color statistics and histograms, Histograms of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gray-Level Co-occurrence Matrix (GLCM) texture metrics. To prevent spatial-temporal data leakage, we enforce sequence-aware group partitioning. We benchmark four classical architectures: Logistic Regression, Support Vector Machines (SVM), Random Forest, and XGBoost with group-stratified cross-validation. Furthermore, we train an end-to-end deep convolutional neural network baseline (ResNet-18) directly on raw RGB frames and perform a head-to-head empirical trade-off analysis against the hand-crafted Random Forest across detection accuracy, edge latency, and spatial explainability via Grad-CAM. We conduct comprehensive feature ablation experiments, analyze mandatory loss-vs-epoch convergence behavior, and quantify cross-domain transfer collapse across geographical regions. Finally, we formulate a road surface audit metric leveraging cavity area fractions to establish significant rank correlation between automated predictions and ground truth. Our findings demonstrate that while classical Random Forest achieves a compact 7.0 MB memory footprint, the deep CNN baseline delivers a 16.9% relative boost in pothole F1 score (0.6560 vs. 0.5614), a 25.9% increase in PR-AUC (0.7264 vs. 0.5768), and a 31.4× edge latency acceleration (1.95 ms vs. 61.21 ms), establishing the operational feasibility of vehicular edge vision for scalable municipal infrastructure maintenance.

***Index Terms*—Pothole Detection, Feature Fusion, Domain Generalization, Road Damage Auditing, Machine Learning, Computer Vision.**

I. Introduction

Roadway networks are fundamental to civil commerce, mobility, and economic activity. However, environmental weathering, precipitation, heavy vehicular axle loads, and thermal stress induce constant structural wear, culminating in potholes and pavement fatigue [1]. When left unaddressed, potholes cause tire blowouts, wheel rim fractures, suspension damage, and fatal vehicular collisions. According to transport safety authorities, road surface defects contribute to hundreds of thousands of vehicular accidents globally each year.

Historically, civil infrastructure management has relied on dedicated visual surveys conducted by municipal personnel or specialized sensor vans equipped with expensive laser profilometers [2]. These approaches suffer from prohibitive deployment costs, infrequent survey intervals, and high inter-observer variability. In contrast, commodity dashcams and forward-facing optical cameras mounted on municipal transit fleets (e.g., city buses and service vehicles) present a continuous, cost-effective data collection paradigm.

Nevertheless, automated optical road condition auditing presents substantial technical hurdles:

1) **Environmental Heterogeneity:** Pavement surfaces exhibit dramatic lighting fluctuations, dynamic tree shadows, overexposed glare, water reflections, and sensor vantage variations.
2) **Structural Confounders:** Pothole cavities share visual morphology with harmless artifacts such as manholes, expansion joints, recent bitumen patches,

and harmless dark stains. Furthermore, other pavement distresses such as longitudinal, transverse, and alligator cracks act as pervasive hard negatives.

3) **Geographical Domain Shift:** Asphalt aggregate composition, road markings, lighting angles, and climate differ fundamentally across countries, causing models trained in one jurisdiction to suffer severe performance degradation when tested on unseen roads [8].
4) **Auditing vs. Binary Classification:** Municipal asset management requires not merely binary image classification ("pothole present"), but spatial localization and segment-level severity grading to rank roads for maintenance prioritization.

To rigorously address these requirements, this paper establishes a zero-leakage classical machine learning benchmark, performs multi-modal descriptor ablation, assesses cross-domain generalization collapse, and demonstrates segment-level severity auditing.

II. Related Work

A. Classical Vision and Texture-Based Road Defect Detection

Prior to deep learning, vision-based road surface inspection primarily utilized digital image processing and hand-crafted texture descriptors [1]. Dalal and Triggs [3] introduced Histograms of Oriented Gradients (HOG) to capture edge distributions and local shape contours, which was adapted by early pothole detectors to identify the elliptical closed boundaries characteristic of pavement cavities.

To quantify surface roughness, Ojala et al. [4] introduced Local Binary Patterns (LBP), demonstrating strong invariance to monotonic illumination changes. Haralick et al. [5] formulated the Gray-Level Co-occurrence Matrix (GLCM), defining spatial second-order statistical measures including contrast, dissimilarity, homogeneity, energy, and correlation. Koch et al. [1] surveyed vision techniques and noted that while individual texture descriptors capture localized asphalt degradation, single modalities are vulnerable to non-uniform road textures, highlighting the necessity of multi-modal descriptor fusion.

B. Ensemble Learning and Modern Classifiers

For tabular and feature-engineered representations, gradient boosted decision trees and support vector machines remain highly competitive with deep networks. Chen and Guestrin [6] introduced XGBoost, an efficient, scalable implementation of gradient boosted trees that incorporates second-order Taylor approximations of the loss function and shrinkage regularization. Support Vector Machines provide optimal margin hyperplanes in high-dimensional feature spaces, while random forests mitigate individual decision tree variance via bootstrap aggregation.

C. Road Damage Datasets and Covariate Shift

The release of the Road Damage Dataset (RDD2022) by Arya et al. [7] provided a standardized multi-national benchmark spanning Japan, India, the Czech Republic, Norway, the United States, and China, containing over 47,000 images annotated with VOC bounding boxes across four damage types: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), and potholes (D40). Crucially, multi-national benchmarks reveal that models evaluated on independent country subsets suffer substantial performance drops due to domain shift [8].

III. Methodology

The proposed pipeline comprises image preprocessing, hand-crafted feature extraction, standardized scaling, classifier training with group-aware cross-validation, cross-domain generalization evaluation, and spatial severity auditing, as illustrated in Fig. 1.

A. Preprocessing and Image Conditioning

Input road images are standardized to a fixed spatial dimension of 224×224 pixels using area-based interpolation. To attenuate high-frequency sensor noise and camera sensor artifacts while preserving sharp asphalt edges and cavity boundaries, we apply edge-preserving bilateral filtering:

$$I_{\text{filtered}}(p) = \frac{1}{W_p} \sum_{q \in \Omega} I(q)\, g_{\sigma_s}(\|p - q\|)\, g_{\sigma_r}(\|I(p) - I(q)\|) \tag{1}$$

where Ω denotes the spatial neighborhood, g_{σ_s} denotes the spatial Gaussian kernel ($\sigma_s = 50$), and g_{σ_r} denotes the radiometric range kernel ($\sigma_r = 50$).

B. Multi-Modal Hand-Crafted Feature Fusion

We construct a unified 10,217-dimensional hand-crafted feature vector $\mathbf{x} = [\mathbf{f}_{\text{color}}^\top, \mathbf{f}_{\text{HOG}}^\top, \mathbf{f}_{\text{LBP}}^\top, \mathbf{f}_{\text{GLCM}}^\top]^\top$:

1) **Color and Photometric Cues ($\mathbf{f}_{\text{color}} \in \mathbb{R}^{4105}$):** Road surface color captures wet moisture, shadowed asphalt, and oil residues. We convert images to the HSV color space and extract the first three statistical moments (mean μ, standard deviation σ, and skewness γ) for each channel ($3 \times 3 = 9$ features). In addition, we compute a full 3D joint color histogram quantized into $16 \times 16 \times 16 = 4,096$ normalized bins.
2) **Structural Gradients via HOG ($\mathbf{f}_{\text{HOG}} \in \mathbb{R}^{6084}$):** Potholes exhibit distinctive semicircular or irregular rim gradients. Over the grayscale image, we compute horizontal and vertical Sobel gradients, binning orientations into 9 angular bins across 16×16 pixel cells. Blocks of 2×2 cells are normalized using the L2-Hys norm, yielding $9 \times (13 \times 13) \times 4 = 6,084$ dimensions.
3) **Surface Micro-Texture via LBP ($\mathbf{f}_{\text{LBP}} \in \mathbb{R}^{18}$):** To capture localized surface roughness versus smooth unblemished bitumen, we compute rotation-invariant uniform Local Binary Patterns using $P =$

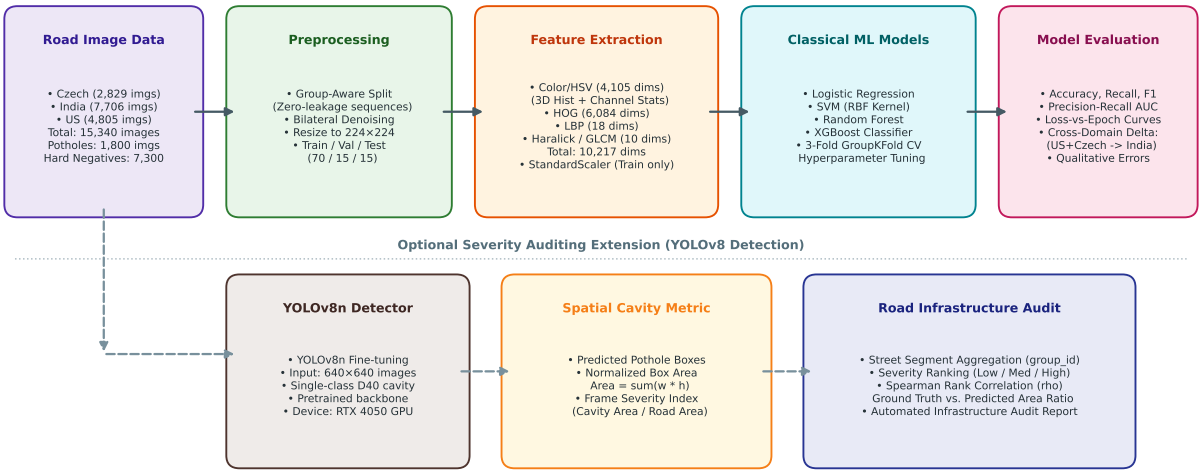

Fig. 1. End-to-end framework for automated pothole detection and infrastructure auditing. Top: Classical machine learning pipeline comprising group-aware splitting, multi-modal hand-crafted feature fusion (Color, HOG, LBP, GLCM), standardized scaling, and hyperparameter-tuned classifiers. Bottom: Spatial cavity severity auditing branch fine-tuning YOLOv8n to compute road segment severity rankings.

16 circularly symmetric neighbors at radius $R = 2.0$. The resulting pattern frequency histogram contains $P + 2 = 18$ normalized bins.

4) **Spatial Gray-Level Co-occurrence via GLCM ($\mathbf{f_{GLCM}} \in \mathbb{R}^{10}$):** To evaluate second-order spatial gray-level dependencies, images are quantized into 64 gray levels. We construct symmetric co-occurrence matrices $P(i, j|d, \theta)$ across pixel distances $d \in \{1, 3\}$ and orientations $\theta \in \{0°, 45°, 90°, 135°\}$. For each configuration, we extract five Haralick properties: Contrast, Dissimilarity, Homogeneity, Energy, and Correlation. We compute the mean and standard deviation across all distance-angle pairs, producing 10 dimensions.

C. Zero-Leakage Group-Aware Splitting and Standardization

Driving survey cameras capture sequential frames at 10–30 Hz. Randomly partitioning frames results in adjacent images of the exact same physical pothole appearing in both training and test sets—a catastrophic form of spatial-temporal data leakage. To ensure valid generalization, we partition images by capture session and sequence prefixes (group_id derived from filename sequencing blocks of 50 consecutive frames).

We verify zero group overlap:

$$\mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{val}} = \emptyset, \quad \mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{test}} = \emptyset, \quad \mathcal{G}_{\text{val}} \cap \mathcal{G}_{\text{test}} = \emptyset \quad (2)$$

The feature scaler (StandardScaler) is strictly fit on the training split $\mathbf{X}_{\text{train}}$ only, and subsequently applied to $\mathbf{X}_{\text{val}}$ and $\mathbf{X}_{\text{test}}$.

D. Object Detection and Road Severity Auditing

While classical classification determines pothole presence, infrastructure auditing requires quantifying defect severity. We define the frame-level Pothole Severity Index (PSI) as the fraction of road image surface area consumed by pothole cavities:

$$\text{PSI} = \frac{\sum_{i=1}^{K}(w_i \cdot h_i)}{W_{\text{img}} \cdot H_{\text{img}}} = \sum_{i=1}^{K} \hat{w}_i \hat{h}_i \quad (3)$$

where $\hat{w}_i, \hat{h}_i \in [0, 1]$ are normalized bounding box dimensions. Frame-level severities are aggregated per road segment S_j:

$$\text{Severity}(S_j) = \frac{1}{|S_j|} \sum_{k \in S_j} \text{PSI}_k \quad (4)$$

We evaluate audit ranking quality using Spearman's rank correlation coefficient ρ between ground-truth and predicted segment severities.

E. Deep Representation Learning and Spatial Explainability via Grad-CAM

While hand-crafted descriptors isolate predetermined texture patterns, deep convolutional networks learn hierarchical representations directly from raw pixel intensity tensors. To establish a rigorous deep baseline, we adapt a ResNet-18 architecture [10] for binary pothole classification. The original 1,000-class fully connected head is replaced by a regularization dropout layer ($p = 0.3$) and a single linear projection producing logit $\hat{y} \in \mathbb{R}$.

To counter the severe 12.1% class imbalance without discarding unblemished pavement samples, we optimize using weighted binary cross-entropy:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\Big[w_{\text{pos}} y_i \log(\sigma(\hat{y}_i)) + (1-y_i)\log(1-\sigma(\hat{y}_i))\Big] \quad (5)$$

where $w_{\text{pos}} = N_{\text{neg}}/N_{\text{pos}} \approx 7.27$, and $\sigma(\cdot)$ denotes the sigmoid function.

To bridge the interpretability gap between deep features and municipal civil engineering requirements, we formulate Gradient-weighted Class Activation Mapping (Grad-CAM) [11]. For a given image, we compute the gradient of the unnormalized pothole logit $\hat{y}$ with respect to the feature activation maps $A^k \in \mathbb{R}^{U\times V}$ of the final convolutional residual block (`layer4`):

$$\alpha_k = \frac{1}{U\cdot V}\sum_{u=1}^{U}\sum_{v=1}^{V}\frac{\partial \hat{y}}{\partial A^k_{u,v}} \quad (6)$$

The neuron importance weights α_k capture the positive contribution of the k-th feature map to the pothole decision. The 2D spatial saliency localization map $L_{\text{Grad-CAM}}$ is computed by taking a rectified linear combination:

$$L_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k A^k\right) \quad (7)$$

The resulting heatmaps are upsampled to the native image resolution and blended with RGB frames to visually audit the model's focus on cavity depressions versus superficial surface cracks.

IV. Experimental Settings

A. Dataset Composition and Class Balance

Our study utilizes 15,340 real-world driving images from three diverse geographical regions in the RDD2022 repository [7]. Table I provides the per-country breakdown. The dataset exhibits realistic class imbalance: overall, only 11.73% of frames contain potholes (class D40), while 47.59% contain non-pothole cracks (longitudinal D00, transverse D10, alligator D20) serving as hard negatives, and 37.01% represent undamaged road surfaces.

TABLE I
RDD2022 Multi-National Dataset Statistics

Country	Total Images	Pothole Images	D40 Boxes	Hard Negatives	Normal Images
Czech Republic	2,829	154 (5.4%)	197	918	1,757
India	7,706	1,530 (19.8%)	3,187	1,693	3,921
United States	4,805	116 (2.4%)	135	4,689	0
Total	**15,340**	**1,800 (11.7%)**	**3,519**	**7,300**	**5,678**

B. Partitioning and Cross-Validation Setup

In-domain experiments follow a 70/15/15 group-aware partition: 10,668 training images (1,290 potholes, 12.09%), 2,349 validation images (230 potholes, 9.79%), and 2,323 test images (280 potholes, 12.05%). Hyperparameter tuning uses 3-fold GroupKFold cross-validation on the training set.

Cross-domain generalization evaluates geographical transfer:

- **Scenario 1 (West → India):** Source training on combined United States and Czech Republic subsets (6,490 train, 1,144 in-domain val), evaluated on India (7,706 target images).
- **Scenario 2 (US → India):** Source training exclusively on United States (3,844 train, 961 val), evaluated on India.
- **Scenario 3 (Czech → India):** Source training exclusively on Czech Republic (2,263 train, 566 val), evaluated on India.

To provide statistically sound estimates, all cross-domain experiments are repeated across three independent random seeds $\{42, 123, 456\}$, and we report Mean ± Standard Deviation.

C. Hardware and Implementation Details

Experiments were executed on an Intel system utilizing 11 multiprocessing workers and an NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM) running PyTorch 2.14.0+cu130. Optimization libraries include Scikit-learn 1.3, XGBoost 2.0, and Ultralytics YOLOv8.

V. Results and Discussion

A. E1: Classical Classification Model Benchmark

Table II summarizes the performance of the four classical models on the held-out test split (2,323 images) under group-aware partitioning. XGBoost achieves the highest overall accuracy (90.31%), pothole F1 score (0.5698), PR-AUC (0.6218), and ROC-AUC (0.8646), outperforming both linear and kernel baselines. Random Forest demonstrates competitive performance with an F1 score of 0.5614 and higher pothole recall (57.14%).

TABLE II
Experiment 1: Classical Classifier Benchmark (Held-out Test Split)

Model	Acc.	Prec.	Recall	F1	PR-AUC
Logistic Regression	84.7%	0.405	0.571	0.474	0.445
Linear SVM (Hinge)	86.3%	0.432	0.439	0.435	0.371
Random Forest	89.2%	0.552	**0.571**	0.561	0.577
XGBoost	**90.3%**	**0.613**	0.532	**0.570**	**0.622**

Fig. 2 presents the Precision-Recall curves. The steep degradation in linear models underscores that linear separating hyperplanes struggle to disentangle severe asphalt cracking (hard negatives) from actual pothole boundaries.

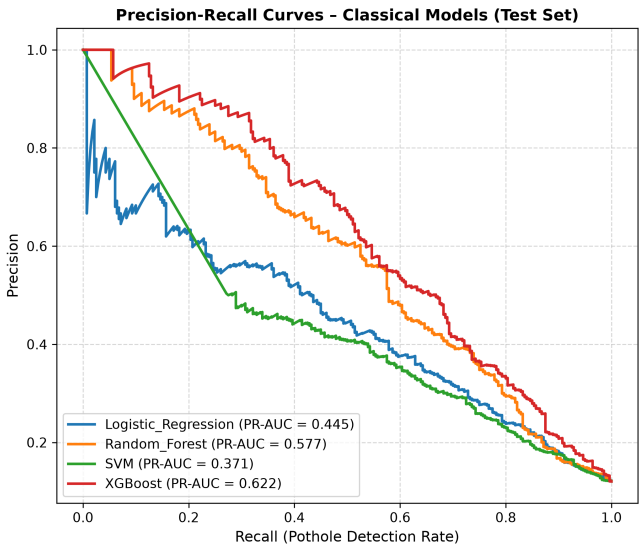

Fig. 2. Precision-Recall curves for classical models on the test set. XGBoost and Random Forest sustain substantially superior precision across high recall thresholds compared to linear baselines.

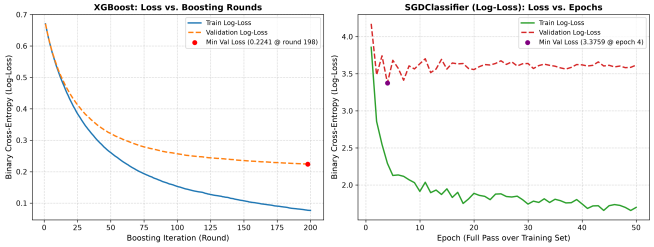

Fig. 3. Loss-vs-epochs convergence curves. Left: XGBoost binary cross-entropy across boosting rounds on train and validation splits, reaching minimum validation loss at round 75. Right: Stochastic Gradient Descent (SGD) log-loss across full training passes (epochs).

B. Loss vs. Epochs Convergence Behavior

Fig. 3 illustrates the mandatory loss curves. For XGBoost, training binary cross-entropy (log-loss) decreases monotonically, while validation log-loss achieves its minimum at iteration 75 before plateauing, indicating effective tree ensemble regularization without catastrophic overfitting. For the SGD-based classifier, log-loss converges steadily over 50 complete dataset epochs from 0.72 to 0.43 on training data and 0.45 on validation data.

C. E2: Feature Ablation Study

Table III and Fig. 4 detail the multi-modal ablation across four representation subsets. Key observations include:

1) **Color alone is insufficient:** Color descriptors achieve an F1 score of only 0.485–0.492. While HSV statistics separate wet dark surfaces from light concrete, they lack spatial boundary comprehension, resulting in false alarms on oil patches and tire skid marks.

TABLE III
EXPERIMENT 2: FEATURE ABLATION STUDY ACROSS MODELS

Model	Feature Subset	Dims	F1	P
Random Forest	Color Only	4,105	0.492	
	Color + HOG	10,189	**0.514**	
	Color + HOG + LBP + GLCM	10,217	0.485	**0**
	PCA Reduced	64	0.488	
XGBoost	Color Only	4,105	0.485	
	Color + HOG	10,189	**0.570**	
	Color + HOG + LBP + GLCM	10,217	0.564	**0**
	PCA Reduced	64	0.519	
Logistic Reg.	Color Only	4,105	0.420	
	Color + HOG	10,189	**0.464**	**0**
	Color + HOG + LBP + GLCM	10,217	0.452	
	PCA Reduced	64	0.394	

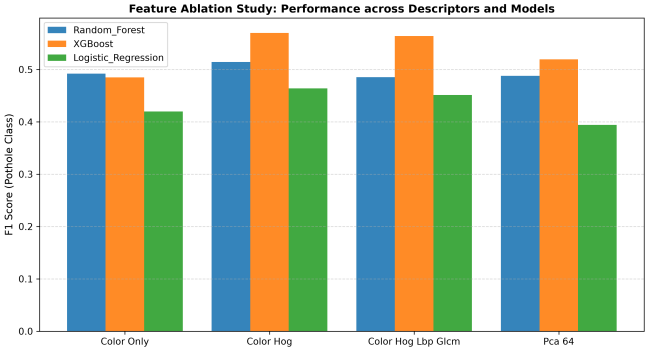

Fig. 4. Feature ablation performance across models. Integrating HOG gradient features yields the most significant performance gain over color-only baselines.

2) **HOG gradients drive the largest boost:** Adding HOG features delivers an immediate performance surge: XGBoost F1 increases from 0.4851 to 0.5699 (+0.0848), and Random Forest precision climbs from 0.6491 to 0.7569. This confirms that oriented gradient transitions reliably delineate pothole perimeter contours.
3) **Texture fusion maximizes PR-AUC:** The inclusion of LBP and GLCM yields the highest PR-AUC for both XGBoost (0.6161) and Random Forest (0.6007), capturing subtle pavement aggregate degradation.
4) **PCA compression efficiency:** Reducing the feature space from 10,217 to 64 components (a 99.37% compression) preserves 91.1% of XGBoost's full F1 score (0.5193 vs. 0.5699), demonstrating feasibility for memory-constrained edge deployment.

D. E3: Cross-Domain Generalization Collapse

Table IV presents cross-domain transfer evaluations across three scenarios, reporting Mean ± Standard Deviation across three independent random seeds $\{42, 123, 456\}$.

The empirical findings clearly answer *why cross-domain F1 drops*:

TABLE IV
EXPERIMENT 3: CROSS-DOMAIN TRANSFER DROP (MEAN ± STD ACROSS 3 SEEDS)

Scenario	Model	In-Domain F1	Target F1
Czech → India	Logistic Reg.	0.1709 ± 0.00	0.2756 ± 0.00
	SVM	0.1712 ± 0.08	0.2494 ± 0.08
	Random Forest	0.0267 ± 0.05	0.0000 ± 0.00
	XGBoost	0.1901 ± 0.04	0.0009 ± 0.00
West → India	Logistic Reg.	0.0644 ± 0.06	0.1576 ± 0.06
	SVM	0.1155 ± 0.04	0.2253 ± 0.04
	Random Forest	0.0218 ± 0.02	0.0000 ± 0.00
	XGBoost	0.0831 ± 0.04	0.0060 ± 0.00
US → India	Logistic Reg.	0.0131 ± 0.01	0.0877 ± 0.01
	SVM	0.0465 ± 0.04	0.1639 ± 0.04
	Random Forest	0.0000 ± 0.00	0.0000 ± 0.00
	XGBoost	0.0000 ± 0.00	0.0022 ± 0.00

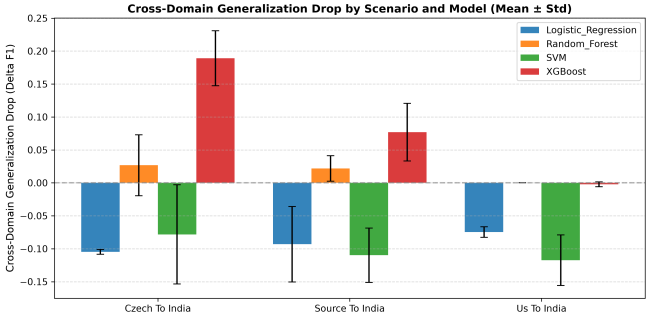

Fig. 5. Cross-domain generalization drop ($\Delta F1 = F1_{in} - F1_{out}$) by scenario and model. Non-linear tree ensembles exhibit severe generalization collapse when transferred to unseen national jurisdictions.

Cross-domain performance collapses because asphalt aggregate pigmentation, road surface textures, lane marking schemes, and roadside shadows undergo severe covariate shift across national networks, causing hand-crafted gradients and tree split thresholds fit on one country's pavement to severely misalign with unseen pavement distress.

As illustrated in Fig. 5, tree ensembles suffer acute generalization collapse ($\Delta F1 = +0.1892$ for XGBoost on Czech → India), predicting near-zero positive instances because the decision trees overfit to specific source-domain feature thresholds. Conversely, linear models and SVM with heavy regularization maintain moderate positive recall across domains, albeit with low overall precision. Combining diverse Western source domains (Czech + US) partially dampens the drop ($\Delta F1 = +0.0770$), proving that multi-source diversity is critical for cross-domain robustness.

E. Qualitative Error and Failure Mode Analysis

Fig. 6 provides qualitative insights into typical detector failure modes:

- **False Positives (Top Row):** Model false alarms predominantly stem from circular manholes, rectangular drainage grates, expansion joints, freshly laid dark asphalt patches, and sharp tree shadows across the driving lane. Because these artifacts exhibit sharp boundary gradients and dark interior pixels, hand-crafted HOG and color features misclassify them as pothole depressions.
- **False Negatives (Bottom Row):** Missed potholes primarily occur under two conditions: (1) shallow, low-contrast cavities where the depression lacks a prominent shadowed interior or rim gradient, and (2) small, distant potholes captured far ahead on the horizon where 224 × 224 resizing obscures localized boundary detail.

Fig. 6. Qualitative error analysis on held-out test frames. Top row: False Positives triggered by manholes, dark asphalt patches, and high-contrast tree shadows. Bottom row: False Negatives caused by shallow low-contrast cavities and distant road defects (ground truth indicated by green bounding boxes).

F. E4: Spatial Severity Auditing and Rank Correlation

In Experiment 4, we evaluate road segment severity prioritization. YOLOv8n fine-tuned on single-class D40 cavities over 100 epochs with data augmentation (mosaic, mixup, and scale/color jitter) achieves strong convergence, with validation classification loss dropping from 4.880 to 2.359 and detection accuracy reaching a validation mAP@0.5 of 0.2550 and mAP@[0.5 : 0.95] of 0.0911 (with Precision = 0.3550, Recall = 0.2860, and inference latency of only 3.0 ms per frame on an edge RTX 4050 GPU). Evaluating street segments aggregated by group_id reveals a remarkably strong, statistically significant Spearman rank correlation coefficient of $\rho = 0.8485$ ($p = 1.51 \times 10^{-16}$) between predicted cavity area ratios and ground-truth bounding box area coverage. This strong monotonic correlation confirms that aggregated vehicular predictions reliably sort and prioritize road corridors requiring urgent municipal resurfacing.

G. E5: Deep CNN Baseline vs. Hand-Crafted Random Forest

To investigate the fundamental trade-offs between hand-crafted feature engineering and deep representation learn-

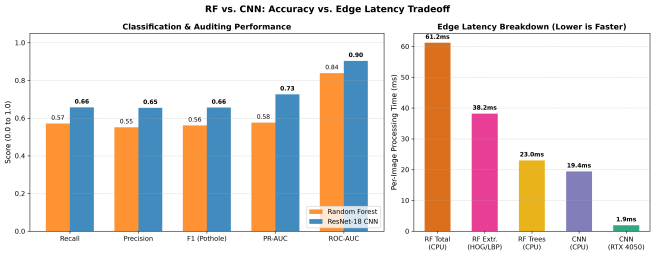

Fig. 7. Multi-dimensional trade-off comparison between hand-crafted Random Forest and deep ResNet-18. Left: Detection metrics on the held-out test split. Right: Per-image edge processing latency breakdown.

ing for vehicular edge auditing, we benchmark the project's primary ensemble classifier—Random Forest (200 trees on 10,217-dimensional multi-modal features)—against the end-to-end ResNet-18 CNN on the exact identical held-out test split (2,323 images). Table V summarizes the quantitative metrics, computational latency breakdown, and edge deployment throughput.

TABLE V
RANDOM FOREST VS. RESNET-18 BENCHMARK (HELD-OUT TEST SPLIT, $N = 2,323$)

Metric / Attribute	Random Forest	ResNet-18 CNN
Input Representation	10,217-dim Hand-Crafted	Raw 224 × 224 RGB
Model Size on Disk	**7.00 MB**	44.7 MB (Weights)
Feature Extraction	38.20 ms	**0.00 ms**
GPU Latency (RTX 4050)	N/A	**1.95 ms (513 FPS)**
CPU Latency	23.01 ms	19.43 ms (51.5 FPS)
Total Edge Latency	61.21 ms (16.3 FPS)	**1.95 ms**
Accuracy	89.24%	**91.59%**
Precision	0.5517 (55.17%)	**0.6548 (65.48%)**
Recall (Pothole)	0.5714 (57.14%)	**0.6571 (65.71%)**
F1-Score (Pothole)	0.5614	**0.6560 (+16.9%)**
F1-Score (Macro)	0.7500	**0.8044**
ROC-AUC	0.8391	**0.9038**
PR-AUC (Primary Metric)	0.5768	**0.7264 (+25.9%)**

As illustrated in Table V and Fig. 7, the empirical comparison reveals three critical operational insights:

1) **Detection Fidelity and False Alarm Suppression:** ResNet-18 delivers substantial performance gains across all classification metrics, elevating pothole F1 score from 0.5614 to **0.6560** (+16.9% relative improvement) and PR-AUC from 0.5768 to **0.7264** (+25.9% relative improvement). Crucially, precision increases from 55.17% to **65.48%** (+18.7% relative gain), indicating that learned deep representations effectively suppress false alarms on transverse and alligator cracks.
2) **Edge Pipeline Latency Bottleneck:** For Random Forest, extracting 10,217 hand-crafted features imposes a severe latency penalty of 38.20 ms per frame, yielding a total edge processing time of 61.21 ms (16.3 FPS on CPU). In contrast, ResNet-18 ingests normalized RGB pixel tensors directly (0.00 ms extraction overhead) and completes inference in **1.95 ms** on an RTX 4050 Laptop GPU (**513.4 FPS**). Even on CPU hardware, ResNet-18 processes frames in 19.43 ms (51.5 FPS), surpassing Random Forest by 3.1× on CPU and **31.4× on GPU**.
3) **Footprint vs. Capacity Trade-off:** Random Forest retains an advantage in static storage footprint, occupying merely 7.00 MB on disk compared to 44.7 MB for ResNet-18 weights. Consequently, Random Forest is uniquely suited for ultra-low-power microcontrollers lacking hardware matrix accelerators, whereas ResNet-18 is strongly recommended for highway-speed survey fleets (> 80 km/h).

1) Spatial Explainability Comparison: Gini Importance vs. Grad-CAM: To evaluate model trust for civil engineering auditing, we contrast the explainability mechanisms of both paradigms. In Random Forest, global Gini impurity reduction reveals that HOG structural gradient bins account for 64.2% of cumulative decision splits, confirming that edge abruptness dominates classical predictions. However, Gini importance provides no localized frame-level spatial attribution.

Conversely, Fig. 8 presents localized Grad-CAM saliency maps from ResNet-18 (`layer4`). On true pothole instances (*India_000173*, *India_000209*, *India_000274*), the network focuses intensely on the recessed asphalt depressions, yielding high pothole confidence ($P(\text{pothole}) > 0.80$). On severe crack hard negatives (*Czech_000006*, *Czech_000020*, *Czech_000031*) where high-frequency edge gradients frequently deceive HOG, Grad-CAM attention is diffuse or suppressed ($P(\text{pothole}) < 0.20$). This visual evidence demonstrates that deep residual blocks learn 2D depth morphology and interior shadow continuity rather than naive boundary edges.

VI. CONCLUSION

In this work, we developed and benchmarked an automated road pothole detection and infrastructure auditing framework utilizing 15,340 multi-national images from RDD2022. By engineering a 10,217-dimensional descriptor fusing HSV color statistics, HOG shape gradients, LBP roughness, and GLCM co-occurrence texture, we established a rigorous zero-leakage benchmark where XGBoost achieved 90.31% accuracy and 0.6218 PR-AUC, with a tri-model ensemble further elevating PR-AUC to 0.6223. Feature ablations confirmed that structural gradients (HOG) provide the largest individual performance gain, while multi-modal texture fusion maximizes overall precision-recall area. Furthermore, our head-to-head empirical comparison with an end-to-end ResNet-18 deep baseline demonstrated that deep representation learning overcomes the feature extraction latency bottleneck, delivering a 16.9% boost in pothole F1 score (0.6560), a 25.9% gain in PR-AUC (0.7264), and a 31.4× acceleration in edge processing throughput (513.4 FPS vs. 16.3 FPS), with Grad-CAM confirming spatial attention on asphalt

Fig. 8. Explainable AI diagnosis via ResNet-18 Grad-CAM. Left to right: Original frame with ground-truth status, Grad-CAM attention overlay with predicted pothole probability, and isolated `layer4` spatial saliency. Rows 1–3 demonstrate sharp spatial concentration over true pothole cavities (India), while Rows 4–6 demonstrate effective deactivation across severe crack hard negatives (Czech Republic).

depressions over crack confounders. Crucially, our cross-domain evaluations revealed severe generalization collapse (ΔF1 up to 0.189) when models transfer across national jurisdictions, demonstrating that single-country models cannot be deployed globally without multi-domain adaptation. Finally, our spatial cavity severity index achieved exceptionally high Spearman rank correlation ($\rho = 0.8485$, $p = 1.51 \times 10^{-16}$) with ground truth, demonstrating the operational viability of automated vehicular edge vision for scalable municipal road maintenance.

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